Lesson 1: Time Math: Hours, Minutes and Timesheets

Assignment

1. Complete the table. Write fractions in simplest form. Round decimals to 2 places.

Minutes	30	15	45	20	12	50
Fraction of an hour						
Decimal hours						

2. Change to decimal hours. Round to 2 decimal places if needed.

(a) 4 h 30 min (b) 6 h 45 min (c) 2 h 10 min

(d) 7 h 24 min (e) 9 h 5 min (f) 135 min

3. Change to hours and minutes.

(a) 3.25 h (b) 5.5 h (c) 1.8 h (d) 6.1 h (e) 0.6 h (f) 8.95 h

4. Kai's work app says he worked 6.3 h on Saturday. He writes "6 h 30 min" on his timesheet. What mistake did he make? What should he write?

5. Write each time as a mixed number of hours.

(a) 3 h 15 min (b) 5 h 20 min

Write each as a mixed number of hours, then in hours and minutes.

(c) $\frac{9}{2}$ h (d) $\frac{10}{3}$ h

(e) A rink rents ice time by the quarter hour. A hockey team books $3\frac{3}{4}$ h. Write $3\frac{3}{4}$ as an improper fraction. How

many quarter hours is that?

6. Add or subtract. Regroup so the minutes are less than 60.

(a) 5 h 20 min + 3 h 25 min

(b) 6 h 45 min + 2 h 35 min

(c) 4 h 50 min + 3 h 40 min + 1 h 55 min

(d) 9 h 10 min − 4 h 35 min

(e) 12 h − 7 h 40 min

7. A mechanic records the time for each job on a car: oil change 45 min, brake job 2 h 20 min, tire swap 1 h 5 min.

(a) What is the total time in hours and minutes?

(b) The shop bills in decimal hours. Write the total in decimal hours, to 2 decimal places.

8. Complete the table.

12-hour clock	7:20 a.m.	2:45 p.m.	12:15 a.m.	11:30 p.m.	12:50 p.m.
24-hour clock					

24-hour clock	06:05	13:35	00:10	18:00	22:40
12-hour clock					

9. Find the elapsed time.

(a) 9:20 a.m. to 11:50 a.m.

(b) 7:45 a.m. to 1:15 p.m.

(c) 11:25 a.m. to 4:05 p.m.

(d) 08:40 to 16:25

(e) 13:55 to 21:10

10. These times go past midnight. Find the elapsed time.

(a) 9:30 p.m. to 2:15 a.m.

(b) 23:10 to 06:40

(c) A nurse at the Foothills hospital works a night shift from 19:00 to 07:30 the next morning. How long is the shift, in hours and minutes and in decimal hours?

11. Find the end time.

(a) A turkey goes into the oven at 11:40 a.m. It needs 3 h 45 min to cook. When will it be done?

(b) A warehouse shift starts at 15:30 and lasts 8 h 45 min. When does it end? Give the time in both clocks.

12. Find the hours worked. Give each answer in hours and minutes and in decimal hours (to 2 decimal places).

(a) Leo works at a hardware store from 8:45 a.m. to 5:00 p.m., with a 30 min unpaid lunch.

(b) A server works a split shift: 11:00 a.m. to 2:15 p.m., and 5:00 p.m. to 9:40 p.m.

13. Mei is an apprentice electrician. Her timesheet uses the 24-hour clock. Complete it.

Day	**In**	**Out**	**Time in to out**	**Unpaid break**	**Hours worked**
Monday	06:30	15:00		30 min	
Tuesday	06:30	14:45		30 min	
Wednesday	07:15	15:50		45 min	
Thursday	12:00	20:20		30 min	
Friday	13:40	19:05		30 min	
				Total hours (hours and minutes)	
				Total hours (decimal hours, 2 places)	

14. Ana worked from 9:50 a.m. to 4:20 p.m. with no break. Here is her work:
4:20 p.m. = 16:20 $16.20 - 9.50 = 6.70$ "I worked 6.70 h."
Find her mistake. Then find the correct time worked, in hours and minutes and in decimal hours.

15. Jesse wants to work exactly 20 h this week. So far he has worked 3 h 45 min on Monday, 4 h 20 min on Tuesday and 5 h 30 min on Thursday. His Saturday shift starts at 10:15 a.m. and will include a 30 min unpaid break. What time should his Saturday shift end?

16. 3 h 45 min written in decimal hours is
A. 3.45 h **B.** 3.75 h **C.** 0.75 h **D.** 225 h

17. A shift from 9:40 p.m. to 6:10 a.m. lasts
A. 3 h 30 min **B.** 8 h 30 min **C.** 9 h 30 min **D.** 15 h 30 min

18. Numerical Response 5 h 42 min written in decimal hours is _____ h.

19. Numerical Response Dana works Saturday and Sunday, from 7:00 a.m. to 3:15 p.m. each day, with a 30 min unpaid break each day. Her total for the weekend, in decimal hours, is _____ h.

Lesson 2: Wages and Salaries: Gross Pay and Pay Periods

Assignment

1. Fill in each blank with a word from the box.

wage	salary	gross pay	net pay	biweekly	semi-monthly	minimum wage

(a) Pay of a set amount for each hour worked is a _____.

(b) A set amount paid for a whole year is a _____.

(c) Everything you earn before deductions is your _____.

(d) Being paid every two weeks, 26 times a year, is being paid _____.

(e) Being paid twice a month, 24 times a year, is being paid _____.

(f) The lowest hourly wage an employer may legally pay is the _____.

(g) Name one job that usually pays a wage: _____
and one job that usually pays a salary: _____

2. Find the gross pay. Round to the nearest cent.

	Time worked	**Hourly wage**	**Hours as a decimal**	**Gross pay**
(a)	8 h	$15.00		
(b)	5 h	$18.40		
(c)	6 h 30 min	$16.70		
(d)	3 h 45 min	$21.60		
(e)	7 h 15 min	$19.85		

3. Kalani is an apprentice plumber. She earns $22.35/h and works 8 hours a day, Monday to Friday. Estimate her weekly gross pay, then calculate it.

4. Lucas is a landscaper in Okotoks. He earns $18.90/h. Last week he worked 7 h 30 min on Monday, 8 h on Tuesday, 6 h 15 min on Wednesday, 7 h 45 min on Thursday and 4 h 30 min on Friday. What was his gross pay?

5. Grace is a hospital porter in Medicine Hat. Her hospital uses the 24-hour clock. She earns $20.40/h. Complete her timesheet and find her gross pay.

Day	**Start**	**End**	**Unpaid break**	**Hours paid**
Monday	07:00	15:30	30 min	
Tuesday	07:00	15:30	30 min	
Thursday	13:00	19:15	30 min	
Friday	15:00	23:00	30 min	
Saturday	09:00	13:30	none	
			Total hours	

6. Find the hourly wage. Check each answer by multiplying.

(a) $510.00 for 34 hours (b) $364.80 for 19 hours (c) $657.00 for 36 hours

7. Is each worker in Alberta paid at least minimum wage? Use the rules from Section 3.

Worker	**Hourly wage**	**Minimum**	**Legal?**
(a) Aiden, 19: $435.00 for 30 h			
(b) Mia, 16, in March: $297.00 for 22 h			
(c) Mia, in August: $405.00 for 30 h			
(d) Omar, 45: $612.00 for 36 h			

8. Kofi is a payroll clerk with a salary of $63 960 a year. Complete the table.

Paid	weekly	biweekly	semi-monthly	monthly
Pay per cheque				

9. Find each person's yearly salary. Who earns the most in a year?

(a) Dana is paid $1987.50 semi-monthly.

(b) Felix is paid $865.40 weekly.

(c) Gwen is paid $3850.00 monthly.

10. (a) Jordan is a youth worker with a salary of \$44 460 a year. He works 38 hours a week. What is the hourly equivalent of his salary?

(b) Tess works at a hair salon for \$17.80/h, 25 hours a week. She works 48 weeks a year (she takes 4 weeks off, unpaid). What is her yearly gross pay?

11. Malik is offered two jobs at a distribution centre. **Offer A** pays \$19.60/h for 30 hours a week. **Offer B** pays a salary of \$33 800 a year for 35 hours a week.

(a) Find the weekly gross pay for each offer.

(b) Find the pay per hour for Offer B, to the nearest cent. Which offer pays more per hour?

(c) Which offer would you advise Malik to take? Give a reason about money and one other reason.

12. Ravi's salary is \$41 600 a year and he is paid biweekly. He says, "I get two cheques a month, so each cheque is $\$41\,600 \div 12 \div 2 = \$1733.33$." Explain his mistake and find the correct amount of each cheque.

13. Nadia works at a pet store for \$16.50/h, 24 hours a week. Her manager gives her a choice: a raise to \$17.75/h for the same hours, *or* 3 more hours a week at \$16.50/h.

(a) Which choice gives her more weekly pay, and by how much?

(b) Give one reason Nadia might still choose the raise.

14. Leah and Sam both earn a salary of \$57 200 a year. Leah is paid biweekly. Sam is paid semi-monthly.

(a) Find the amount of each person's paycheque.

(b) Who earns more in a year? Explain why Leah's cheques are smaller.

15. A salary of $41 600 a year for a 40-hour week works out to an hourly equivalent of

A. $800.00 **B.** $20.00 **C.** $1040.00 **D.** $86.67

16. The gross pay for 7 h 30 min of work at $16.00/h is

A. $116.80 **B.** $120.00 **C.** $112.00 **D.** $128.00

17. Numerical Response A salary of $51 000 a year is paid semi-monthly. Each paycheque is $_____.

18. Numerical Response Hamid works 6 h 15 min at $18.40/h. His gross pay, to the nearest dollar, is $_____.

Lesson 3: Overtime: Time-and-a-Half and Double Time

Assignment

1. Complete the table.

Base wage	\$15.00/h	\$17.40/h	\$20.60/h	\$32.00/h	\$13.00/h
Time and a half					
Double time					

2. Find the overtime pay.

(a) 3 h at time and a half; base wage \$16.00/h

(b) 5.5 h at time and a half; base wage \$21.40/h

(c) 4 h at double time; base wage \$18.75/h

(d) 2.25 h at time and a half; base wage \$24.80/h

3. Shay works at a grocery store for \$16.40/h. Last week she worked 40 regular hours and 3.5 overtime hours at time and a half.

(a) Find her regular pay.

(b) Find her overtime pay.

(c) Find her gross pay.

4. A call centre pays time and a half for every hour over 37.5 in a week. Aiden earns \$18.80/h. This week he worked 42 hours. Find his gross pay.

5. Hannah earns \$20.00/h. One week she worked 40 regular hours and 5 overtime hours at time and a half. Two friends found her gross pay. Each made a mistake. Explain each mistake, then find the correct gross pay.

Ben: overtime rate = \$20.00 + \$1.50 = \$21.50
$40 \times \$20.00 + 5 \times \$21.50 = \$907.50$

Chloe: $45 \times \$30.00 = \$1350.00$

6. Lena is a health care aide. She earns \$23.50/h. Her union agreement pays double time on general holidays. She works an 8 h shift on Christmas Day.

(a) How much does she earn for the shift?

(b) How much more is that than a regular 8 h shift?

7. Darnell works on a drilling rig for \$31.00/h. His contract pays time and a half after 8 hours in a day and double time after 12 hours in a day. One week he had 44 regular hours, 6 hours at time and a half and 2 hours at double time. Find his gross pay.

8. Nadia works at a vet clinic in Okotoks for \$17.20/h. Use the Alberta 8/44 rule.

Day	**Hours**	**Hours over 8**
Monday	9	
Tuesday	8	
Wednesday	11	
Thursday	6.5	
Friday	8.75	
Total		

(a) Complete the table.
(b) How many overtime hours did she work? Explain.

(c) Find her gross pay.

9. Ravi works at a car wash for \$15.60/h. One week he worked 8 hours a day for 6 days. Use the Alberta 8/44 rule to find his overtime hours and his gross pay.

10. Use the Alberta 8/44 rule to complete the table.

Worker	Hours each day	Total hours	Daily overtime	Weekly overtime	Overtime hours	Regular hours
(a) Owen	10, 10, 9, 8, 6, 4					
(b) Sofia	7.5, 7.5, 7.5, 7.5, 7.5, 7.5					
(c) Mateo	12, 12, 12, 4					
(d) Grace	9, 9, 9, 9, 9, 3					

11. Kiera is an apprentice carpenter. She earns \$18.40/h. Her timesheet uses the 24-hour clock.

Day	Start	End	Unpaid break	Hours worked	Hours over 8
Monday	07:00	17:00	30 min		
Tuesday	07:00	15:30	30 min		
Wednesday	06:00	17:30	1 h		
Thursday	07:00	14:00	30 min		
Friday	07:00	16:45	45 min		
Saturday	08:00	12:00	none		
			Total		

(a) Complete the timesheet.

(b) Use the Alberta 8/44 rule to find her overtime hours.

(c) Find her gross pay.

12. Jun is paid every two weeks (biweekly) at \$17.00/h. In week 1 he worked 48 hours. In week 2 he worked 42 hours. No day was longer than 8 hours. In Alberta, weekly overtime is counted for each week on its own.

(a) Find his overtime hours in each week.

(b) Find his gross pay for the two weeks.

(c) His boss added the two weeks first: $90 - 88 = 2$ overtime hours. How much would Jun lose with the boss's way?

13. Two jobs at a lumber yard. Job A pays \$20.00/h. It is 8 hours a day, 6 days a week, with Alberta overtime. Job B pays \$22.50/h for 40 hours a week (8 hours a day, 5 days).

(a) Which job pays more per week? By how much?

(b) Give one advantage and one disadvantage of working a lot of overtime.

14. Sam earns \$18.00/h. Every week he works 44 regular hours. How many hours of overtime at time and a half must he work to earn \$900.00 in a week?

15. A worker's base wage is \$18.00/h. The time-and-a-half rate is

A. \$9.00/h **B.** \$19.50/h **C.** \$27.00/h **D.** \$36.00/h

16. Lucas works 9 hours a day, Monday to Friday. Under the Alberta 8/44 rule, how many overtime hours does he work?

A. 0 h **B.** 1 h **C.** 5 h **D.** 6 h

17. Numerical Response A worker in Alberta works these hours: Monday 11, Tuesday 7, Wednesday 10.5, Thursday 8, Friday 9, Saturday 6. Using the 8/44 rule, the number of overtime hours is _____.

18. Numerical Response Marco earns \$17.60/h. He works 5.5 hours of overtime at time and a half. His overtime pay, to the nearest dollar, is \$_____.

Lesson 4: Piecework and Contracts

Assignment

1. How is each worker most likely paid? Use: **hourly wage**, **salary**, **piecework** or **contract**.

Worker	**Paid by**	**Worker**	**Paid by**
(a) a roofer who agrees to re-shingle a house for $9800.00		(b) a grocery store cashier	
(c) a worker paid $0.06 for each flyer delivered		(d) a school principal	
(e) a web designer who builds a website for $1500.00		(f) a cherry picker paid by the bucket	

2. Find each worker's pay.

(a) Carmen solders 36 circuit boards at $2.75 each.

(b) Liam plants 1875 trees at $0.17 per tree.

(c) Abdi picks 14 lugs of cherries at $6.50 per lug.

(d) Grace writes an 850-word article at $0.15 per word.

3. Ravi works at a bike shop on Saturdays and is paid for each repair he finishes. Complete the table.

Repair	**Rate**	**Number**	**Work**	**Pay**
Fix a flat tire	$12.00	9		
Tune-up	$45.00	3		
Adjust brakes	$18.00	5		
			Total for Saturday	

4. Sofia sews tote bags for a craft company. She is paid $7.25 for each bag.

(a) On Monday she sews 22 bags in 7.5 hours. Find her pay and her effective hourly rate.

(b) On Tuesday her machine breaks for part of the day, and she sews only 14 bags in 7.5 hours. Find her pay and her effective hourly rate.

(c) Alberta's minimum wage is $15.00 per hour. How much must the company add to Sofia's Tuesday pay?

5. Work backwards. Check each answer by multiplying.

(a) Kenji is paid \$171.00 to put together 38 gift baskets. What is he paid per basket?

(b) Noor does data entry for \$0.85 per form. She earned \$68.00. How many forms did she enter?

(c) Riya plants trees for \$0.22 per tree. How many trees must she plant to earn at least \$400.00 in a day?

6. A moving company pays its packers \$3.75 for each box they pack. It charges customers \$9.50 for each box packed. A packer packs 86 boxes for one customer.

(a) How much does the packer earn?

(b) How much does the company charge the customer for packing?

(c) How much is left for the company's other costs and profit?

7. Tanvir is a handyman. He agrees to build a backyard deck for \$4800.00. The lumber and screws cost \$2135.60. He hires a helper for 16 hours at \$22.00 per hour. Tanvir works 30 hours on the deck himself.

(a) How much does Tanvir pay his helper?

(b) How much does Tanvir earn for himself?

(c) What is Tanvir's effective hourly rate?

8. Jess is a web designer. She agrees to build a website for a bakery for \$1500.00. She has no material costs.

(a) She plans for the job to take 25 hours. What would her effective hourly rate be?

(b) The bakery keeps asking for changes, and the job takes 40 hours. What is her effective hourly rate now?

(c) What could Jess write in her next contract to protect herself?

9. Bianca is bidding on a job to build a wooden fence. The boards, posts and concrete cost \$1640.00. She will hire 2 helpers for 9 hours each at \$21.00 per hour. She wants to earn \$700.00 for herself.

(a) What price should Bianca bid?

(b) Bianca rounds her bid up to the next \$50.00. What does she bid? How much will she earn then?

10. Hiro bids on a contract to clear snow from a strip mall parking lot for the winter. He expects to pay $2350.00 for fuel and salt, and $1230.00 for a helper. He adds 25% to his costs for his profit.

(a) What are Hiro's total costs?

(b) What is his bid?

(c) How much profit will Hiro make if his costs are what he expects?

(d) It is a very snowy winter, and the fuel and salt cost $3100.00 instead. How much profit does he make now?

11. A tile setter can tile a bathroom at $32.00 per hour, or for a contract price of $1250.00. With the contract, she pays $110.00 for the grout and glue.

(a) If the job takes 30 hours, which way pays more, and by how much?

(b) For about how many hours do the two ways pay the same? Round to one decimal place.

(c) The tile setter is new to this kind of tile and is not sure how long it will take. Which way would you choose? Explain.

12. Finish each sentence.

(a) One advantage of piecework for the worker is _____

(b) One disadvantage of piecework for the worker is _____

(c) One advantage of a contract for the customer is _____

(d) One disadvantage of contract work for the contractor is _____

13. Marco did a renovation contract for $3200.00. Materials cost $860.00. He worked 40 hours. Here is his work:

$3200.00 ÷ 40 = $80.00 "I made $80.00 an hour!"

What mistake did Marco make? Find his real effective hourly rate.

14. Owen bids \$6400.00 to finish a basement bathroom. His materials will cost \$3150.00, and he will pay a helper for 32 hours at \$23.00 per hour. Owen expects to work 45 hours himself.

(a) What will Owen earn, and what is his effective hourly rate? Round to the nearest cent.

(b) The customer says another contractor will do it for 10% less. If Owen lowers his price by 10%, what is his new price, his earnings and his effective hourly rate?

(c) The price dropped by 10%. By what percent did Owen's earnings drop? Why is it more than 10%?

15. Ella sews aprons and is paid \$3.20 per apron. How much does she earn for sewing 45 aprons?

A. \$14.06 **B.** \$48.20 **C.** \$144.00 **D.** \$1440.00

16. A contractor does a job for \$900.00. The materials cost \$180.00. The contractor works 24 hours. What is the contractor's effective hourly rate?

A. \$7.50 **B.** \$30.00 **C.** \$37.50 **D.** \$45.00

17. Numerical Response A tree planter is paid \$0.18 per tree. The number of trees she must plant to earn \$369.00 is _____.

Lesson 5: Commission and Salary Plus Commission

Assignment

1. Complete the table.

Commission rate	6%	12%	2.5%	4.25%
As a decimal				
Commission on $2400 of sales				

2. Find the commission. Round to the nearest cent if needed.

(a) sales $3500.00, rate 5%

(b) sales $12 680.00, rate 7%

(c) sales $945.50, rate 15%

(d) sales $56 900.00, rate 2.75%

3. Tyler sells hot tubs and earns 8% commission. He sells a hot tub for $9799.00.

(a) Estimate his commission.

(b) Calculate his commission.

4. Rachel sells home security systems. She earns $520.00 a week plus 3% of her sales.

(a) Find her gross pay in a week when she sells $9450.00.

(b) Find her gross pay in a slow week when she sells $2100.00.

5. Diego works at a shoe store in Medicine Hat. He earns $16.25/h plus 2% commission on his sales. One week he works 30 h and sells $7840.00 of shoes. Find his gross pay.

6. A store pays 4% on the first \$15 000 of monthly sales and 6% on sales over \$15 000. Find the commission for each month.

(a) sales \$12 000 (b) sales \$15 000 (c) sales \$23 500

7. In Alberta, many real estate offices charge 7% on the first \$100 000 of a home's price and 3% on the rest. A house in Airdrie sells for \$485 000.

(a) Find the total commission.

(b) The commission is split equally between the seller's office and the buyer's office. How much does each office get?

8. Kayla sells gym memberships. She earns \$1800.00 a month plus 10% of her sales over a quota of \$12 000.

(a) Find her gross pay in a month when she sells \$16 500.

(b) In another month she sells \$11 200. What is her gross pay? Explain.

9. Find the commission rate as a percent.

(a) \$540.00 on sales of \$12 000.00 (b) \$93.60 on sales of \$1170.00 (c) \$2125.00 on sales of \$85 000.00

10. A travel agent earns 9% commission on the trips she books. Last month she earned \$1314.00 in commission. What was the total value of the trips she booked? Check your answer.

11. Two car-wash supply companies offer Malik a sales job.
Job A: \$650 a week. **Job B:** \$300 a week plus 7% of his sales.

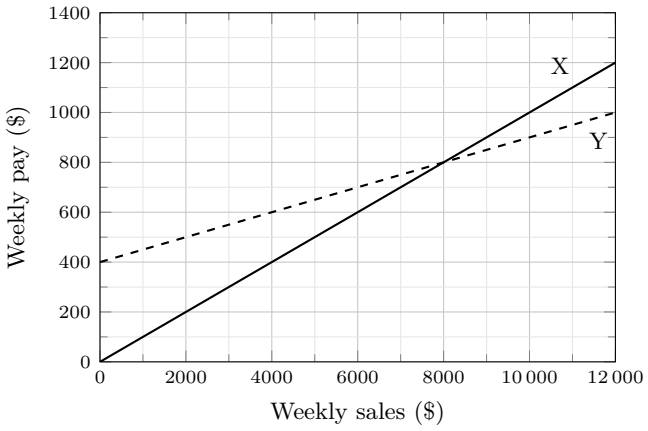
(a) Complete the table.

Weekly sales	\$3000	\$5000	\$7000
Job A pay			
Job B pay			

(b) At what weekly sales do the jobs pay the same?

(c) Malik expects to sell about \$4000 a week. Which job pays more?

12. The graph shows the weekly pay for two plans at a computer store.
Plan X: 10% straight commission. **Plan Y:** \$400 a week plus 5% commission.



(a) Which plan pays more at \$4000 of sales? By how much?

(b) Which plan pays more at \$12 000 of sales?

(c) At what sales do the plans pay the same? What is the pay there?

(d) Why does the line for Plan X start at \$0?

13. Give one advantage and one disadvantage of straight commission for the worker. Then give one reason a store owner might like to pay commission.

14. Sam earns 3% on the first \$10 000 of his sales and 5% on sales over \$10 000. He sold \$18 000. Here is his work:

$$\$18\,000 \times 0.05 = \$900 \quad \text{“My commission is \$900.”}$$

Find his mistake and give the correct commission.

15. A 4% commission on sales of $6250 is

A. $25.00 **B.** $250.00 **C.** $2500.00 **D.** $6500.00

16. Ben earns $400 a week plus 5% commission. In a week when he sells $3000, his gross pay is

A. $150 **B.** $420 **C.** $550 **D.** $1900

17. Numerical Response Leila earned $410.40 in commission on sales of $9120.00. Her commission rate, as a percent, is _____%.

18. Numerical Response A salesperson earns 2% on the first $5000 of sales and 4% on sales over $5000. On sales of $8200, the commission, to the nearest dollar, is $_____.

Lesson 6: Extra Earnings: Tips, Bonuses and Shift Premiums

Assignment

1. Find each tip. Estimate the 15% tip in your head first.

Bill	Estimate of 15%	10% tip	15% tip	20% tip
(a) $30.00				
(b) $48.60				
(c) $112.80				

2. What percent tip did each customer leave? Round to the nearest tenth of a percent.
(a) $7.00 on a $40.00 bill (b) $5.00 on a $36.25 bill (c) $3.50 on a $23.80 taxi ride

3. Tyson is a server in Medicine Hat. He earns $15.00/h. On Saturday he worked 7 hours. His bills totalled $1080.00, and his tips were 17% of the bills.
(a) Find his gross pay for Saturday.

(b) Give one advantage and one disadvantage of a job where a large part of the pay is tips.

4. (a) A hairstylist earned $742.50 last week. She worked 33 hours at $16.50/h, and the rest was tips. How much did she earn in tips?

(b) A bartender's tips one night were $131.10. That was 15% of the bills he served. What was the total of the bills?

5. Four workers at a pizza place share a tip pool of $486.00 by the hours they worked. Complete the table.

Worker	Ana	Ben	Chloe	Dev	**Total**
Hours	7	5	6	9	
Share					

6. Hannah made $264.00 in tips on Friday. She tips out 20% to the kitchen and 5% to the busser.

(a) How much does each get?

(b) How much does Hannah keep?

7. (a) An office worker earns $52 400 a year. Her bonus is 2.5% of her salary. Find her bonus.

(b) A salesperson can choose a bonus of $1200, or 2% of his sales for the year. His sales were $68 500. Which should he choose, and by how much is it better?

8. A furniture store pays a bonus for each sofa sold: $10 for a sofa under $1000, $25 for a sofa from $1000 to $1999, and $40 for a sofa $2000 or more. This week Lucas sold sofas for $899, $1450, $2299, $1999, $750 and $3100. Find his bonus.

9. Kofi works at a gas station in Okotoks. He earns $16.25/h, plus a premium of $1.50/h for every hour after 7:00 p.m. His breaks are paid. Complete the timesheet and find his gross pay for the week.

Day	**Start**	**End**	**Hours worked**	**Hours after 7:00 p.m.**
Monday	3:00 p.m.	11:00 p.m.		
Wednesday	11:00 a.m.	7:00 p.m.		
Thursday	4:00 p.m.	10:00 p.m.		
Saturday	1:30 p.m.	9:30 p.m.		
		Total		

10. A warehouse pays night-shift workers 12% more per hour. The day wage is \$19.75/h.

(a) What is the night-shift hourly rate?

(b) How much does a worker earn for an 8-hour night shift? How much more is that than an 8-hour day shift?

11. (a) A power line worker earns \$46.80/h. For work on live lines he gets danger pay of 15% more per hour. Last week he worked 12 hours on live lines and 28 other hours. Find his gross pay.

(b) A lab worker at a remote camp earns \$4480 for a 14-day rotation. She can choose isolation pay of \$45 per day, or 8% of her pay. Which pays more for one rotation, and by how much?

12. Marco worked 40 hours at \$17.00/h. He gets a night premium of \$2.00/h, and 12 of his hours were at night. Here is how his friend worked out his pay:

$$\$17.00 + \$2.00 = \$19.00$$
$$40 \times \$19.00 = \$760.00$$

(a) What mistake did his friend make?

(b) Find Marco's correct gross pay.

13. Zoe is a bartender at a pub in Canmore. She earns \$15.50/h. This week she worked 10 hours on Saturday and 8 hours on each of four other days. She gets a late-night premium of \$1.25/h, and 15 of her hours were late-night hours. Her share of the tip pool was \$389.40.

(a) How many hours of overtime did she work? (Alberta: over 8 in a day or over 44 in a week, whichever is more.)

(b) Complete her pay summary.

Type of pay	**Work**	**Amount**
Regular pay		
Overtime pay		
Late-night premium		
Tip pool share		\$389.40
Gross pay		

14. Kai worked 6 hours on Sunday. His restaurant shares the tip pool by hours, and his share was \$97.50. The whole pool was \$422.50. How many hours did all the staff work in total?

15. A 15% tip on a \$60.00 bill is

A. \$0.90 **B.** \$6.00 **C.** \$9.00 **D.** \$69.00

16. Ali earns \$18.00/h and gets a \$2.00/h premium for hours after 8:00 p.m. He works an 8-hour shift, and 3 of the hours are after 8:00 p.m. His pay for the shift is

A. \$146.00 **B.** \$150.00 **C.** \$160.00 **D.** \$204.00

17. Numerical Response A carpenter earns \$48 200 a year. His bonus is 3.5% of his salary. His bonus, to the nearest dollar, is \$_____.

18. Numerical Response A customer leaves an \$11.50 tip on a \$68.00 bill. The tip percent, to the nearest tenth of a percent, is _____%.

Lesson 7: Deductions and Net Pay: Reading a Pay Stub

Assignment

1. Fill in each blank with one of these: *gross pay, net pay, deduction, pay stub, taxable income.*

(a) Money your employer takes off your pay is a _____.

(b) The money that goes into your bank account is your _____.

(c) Everything you earned before anything is taken off is your _____.

(d) Gross pay minus before-tax deductions is your _____.

(e) The paper (or web page) that lists your pay and deductions is a _____.

(f) Explain in your own words why your net pay is less than your gross pay.

2. Complete the table.

	Gross pay	**Deductions**	**Total deductions**	**Net pay**
(a)	$450.00	total deductions given	$63.27	
(b)	$1125.50	CPP $66.97, EI $18.46, income tax $157.57		
(c)	$2316.00	CPP $137.80, EI $37.98, income tax $393.72, health plan $45.00		

3. Change each rate to a decimal, then find the deduction. Round to the nearest cent.

(a) CPP at 5.95% of $520.00

(b) EI at 1.64% of $880.00

(c) income tax at 11% of $1360.00

(d) union dues at 2.25% of $1540.00

4. Mia's gross pay is $800.00. Her pay stub shows EI of $131.20 at a rate of 1.64%. Without working out the exact answer, explain why this cannot be right. Then find the correct EI.

5. Owen is a lifeguard at a Calgary rec centre. He earns $16.40 an hour and works 22.5 hours this week. His deductions are CPP 5.95%, EI 1.64% and income tax 8%.

(a) Find his gross pay.

(b) Find each deduction and his net pay.

6. Brooke is a dental assistant in Red Deer. Her gross pay for two weeks is $1875.00. Her deductions are CPP 5.95%, EI 1.64%, income tax 17%, a health and dental plan of $32.40, and a $10.00 donation to the children's hospital. Find her net pay.

7. Dana works at a ski rental shop in Canmore. Overtime is paid at time and a half. Complete her pay stub.

Three Sisters Ski & Board, Canmore, AB						
Employee: Dana Kowalski			Pay period: Dec. 14–20, 2026			
Earnings	**Hours**	**Rate**	**Amount**	**Deductions**	**Rate**	**Amount**
Regular	40	$21.50		CPP	5.95%	
Overtime	6			EI	1.64%	
				Income tax	16%	
				Parking		$15.00
		Gross pay		**Total deductions**		
					Net pay	

8. Tyler's gross pay is $1450.00 and his net pay is $1093.30.

(a) What are his total deductions?

(b) What percent of his gross pay are his deductions?

(c) What percent of his gross pay does he take home?

9. Find each rate as a percent.

(a) EI of \$12.30 on gross pay of \$750.00

(b) income tax of \$214.20 on taxable income of \$1260.00

(c) union dues of \$27.30 on gross pay of \$1820.00

10. Jasmine is a health care aide in Medicine Hat. Her gross pay for two weeks is \$2240.00. She puts 4% of her gross pay into the company pension plan (a before-tax deduction). Her other deductions are CPP 5.95%, EI 1.64%, and income tax 19% of her taxable income.

(a) Find her pension deduction and her taxable income.

(b) Find her CPP, EI and income tax.

(c) Find her total deductions and net pay.

11. Check each line of Jun's pay stub. Exactly one amount is wrong. Find it, fix it, and say whether the mistake would pay Jun too much or too little.

Chinook Garden Market, Calgary, AB

Employee: Jun Park Pay period: Sept. 7–13, 2026

Earnings	**Hours**	**Rate**	**Amount**	**Deductions**	**Rate**	**Amount**
Regular	32	\$18.25	\$584.00	CPP	5.95%	\$34.75
				EI	1.64%	\$9.58
				Income tax	13%	\$75.92
				Health plan		\$9.50
		Gross pay	\$584.00	**Total deductions**		\$129.75
					Net pay	\$545.25

12. Dev's gross pay is \$900.00. His deductions are CPP 5.95%, EI 1.64% and income tax 15%. He has no before-tax deductions. Here is his work:

$$\text{CPP} = 900 \times 0.0595 = 53.55 \qquad \text{EI} = 900 \times 0.0164 = 14.76$$

$$\text{income tax} = (900 - 53.55 - 14.76) \times 0.15 = 831.69 \times 0.15 = 124.75$$

$$\text{net pay} = 900 - 124.75 = \$775.25$$

Find his two mistakes, and find his correct net pay.

13. What if? Kayla works 30 hours a week at $17.00 an hour. Her boss gives her a raise to $18.50 an hour. Her deductions are CPP 5.95%, EI 1.64% and income tax 12%.

(a) How much more is her weekly gross pay after the raise?

(b) Find her weekly net pay before and after the raise.

(c) Her take-home pay goes up by less than $45.00. By how much, and why?

14. Which of these is *net pay*?

A. the pay you earn before anything is taken off

B. the total of all your deductions

C. the pay you get after all deductions are taken off

D. your pay after only the income tax is taken off

15. The EI at 1.64% on gross pay of $640.00 is

A. $1.05 **B.** $10.50 **C.** $104.96 **D.** $629.50

16. Rosa's gross pay is $1200.00. Her union dues are $30.00 (before tax). Her income tax is 20% of her taxable income. Her income tax is

A. $6.00 **B.** $234.00 **C.** $240.00 **D.** $936.00

17. Numerical Response The EI at 1.64% on gross pay of $545.00, to the nearest cent, is $_____.

18. Numerical Response Leo's gross pay is $860.00 and his total deductions are $197.80. The percent of his gross pay that he takes home, to the nearest tenth of a percent, is _____%.

Lesson 8: Working Backwards: How Many Hours Do I Need?

Assignment

1. Complete the table. Show the formula you used.

	Rate ($/h)	**Hours**	**Pay**	**Formula used**
(a)	$15.00	20		
(b)	$16.00		$480.00	
(c)		30	$547.50	
(d)	$22.40		$627.20	
(e)		37.5	$731.25	

2. Fill in the blanks to rearrange each formula.

(a) pay = rate × hours, so hours = _____ and rate = _____

(b) commission = rate × sales, so sales = _____

3. Ella is 16 and works part-time at a bakery during the school year. She earns $13.00 an hour (the Alberta student rate, as of 2026; rates can change).

(a) How many hours must she work to earn $195.00?

(b) How many whole hours must she work to earn at least $230.00?

4. Jordan earns $16.00 an hour and wants to earn $400.00. Here is his work:

hours = $16 \div 400 = 0.04$, so I need 0.04 hours.

Explain his mistake and find the correct answer. Check it.

5. Amara works 5-hour shifts at a coffee shop for $15.75 an hour. She wants to earn at least $1000.00 (gross) for a school trip.

(a) How many whole hours must she work?

(b) How many shifts is that? Check your answer.

6. What if? Noor works 24 hours a week at a pet store for $17.00 an hour.

(a) What is her gross pay for a week?

(b) Noor wants to earn $500.00 a week. How many whole hours a week must she work?

(c) What if she gets a raise to $18.50 an hour? Now how many hours does she need for $500.00?

(d) What hourly wage would give her $500.00 for her usual 24 hours? Round up to the cent.

7. A tree planter in northern Alberta is paid $0.17 for each tree she plants.

(a) How many trees must she plant to earn $340.00 in a day?

(b) One rainy day she plants 1600 trees in 8 hours. What is her pay for the day, and what does that work out to per hour?

8. Bryn works 8 hours a day, Monday to Friday, at a sawmill for $19.00 an hour. Extra hours added to a shift are overtime at time-and-a-half. How many overtime hours does Bryn need to earn $931.00 this week? Check your answer.

9. Find each hourly wage from the pay stub information.

(a) 40 regular hours and 4 overtime hours at time-and-a-half; gross pay $1035.00

(b) 32 regular hours and 8 hours on a statutory holiday, which her employer pays at double time; gross pay $1008.00

10. (a) A real estate assistant earns 5% commission. How much must she sell to earn \$1750.00 in commission?

(b) A car stereo installer earns \$2200.00 a month plus 3% commission on his sales. How much must he sell to earn \$3400.00 this month?

11. Mateo works at a sporting goods store. He earns \$500.00 a week plus 6% commission on his sales *over* a quota of \$8000.00. How much must he sell in total to earn \$800.00 this week?

12. Zoe has two job offers selling solar panels. **Offer A:** 7% commission only. **Offer B:** \$1500.00 a month plus 3% commission.

(a) Complete the table.

Monthly sales	**Offer A pay**	**Offer B pay**
\$30 000		
\$50 000		

(b) Offer A pays 4% more commission than Offer B ($7\% - 3\% = 4\%$). Offer B has a \$1500 salary. At what sales do the two offers pay the same? (Hint: 4% of the sales must make up the \$1500.) Check your answer.

(c) Zoe expects to sell about \$45 000 a month. Which offer should she take? Give one other reason someone might pick the other offer.

13. Jaylen earns \$17.50 an hour. His deductions take 18% of his gross pay. How many hours must he work to take home \$574.00?

14. Harper earns $21.00 an hour as an apprentice electrician. Her deductions are CPP 5.95%, EI 1.64% and income tax 15%. She wants to save $4500.00 for a used truck.

(a) What is her total deduction rate? What percent of her pay does she keep?

(b) Find her net pay for one hour.

(c) How many whole hours must she work to save $4500.00 (if she saves all her net pay)?

(d) She works 20 hours a week. How many weeks will it take?

15. A salesperson earns 4% commission. To earn $1000.00 in commission, she must sell

A. $40.00 **B.** $4000.00 **C.** $25 000.00 **D.** $250 000.00

16. Tariq earns $15.50 an hour. The number of whole hours he must work to earn at least $300.00 is

A. 19 **B.** 19.35 **C.** 20 **D.** 4650

17. Numerical Response Keira earns $18.40 an hour. Her gross pay last week was $598.00. The number of hours she worked, to the nearest tenth, is _____.

18. Numerical Response Luis earns $400.00 a week plus 5% commission on all his sales. To earn $650.00 in one week, his sales must be $_____.

Lesson 9: Unit 2 Review

Assignment

This review is set up like the unit test. Show your work on the written questions. Round money to the nearest cent. Deduction rates are examples.

Part A: Written Response

1. Change each time.

(a) 45 min to decimal hours

(b) 3 h 20 min to decimal hours

(c) 5.6 h to hours and minutes

(d) Add: 7 h 50 min + 5 h 25 min

2. Ella is 16. She works at a pizza place for $15.40/h. Here is her timesheet.

Day	Start	End	Unpaid break	Hours worked
Monday	4:00 p.m.	9:30 p.m.	none	
Wednesday	3:45 p.m.	8:15 p.m.	none	
Saturday	9:00 a.m.	5:30 p.m.	30 min	
Sunday	11:15 a.m.	4:45 p.m.	30 min	
			Total	

(a) Complete the timesheet.

(b) What is her gross pay for the week?

(c) Ella's friend Mo is 17. During the school year he works 20 h a week at a store for $12.50/h. Is that allowed in Alberta? Explain.

3. Ravi's salary as an apprentice welder is $52 520 a year. He works 40 h a week.

(a) How much is each pay if he is paid monthly?

(b) How much is each pay if he is paid biweekly?

(c) What is his hourly equivalent?

(d) Another shop offers him $26.00/h for 37.5 h a week. Which job pays more in a year? Show your work.

4. Marcus is a carpenter in Lethbridge. He earns $21.60/h, with time and a half for overtime under the Alberta rules.

Day	Mon	Tue	Wed	Thu	Fri	**Total**
Hours worked	9	10.5	8	8	9	
Hours over 8						

(a) Complete the table. How many overtime hours does he get? Explain.

(b) Find his gross pay for the week.

5. Some employers pay **double time**.

(a) Jenna earns $17.80/h at a hotel. She works 7.5 h on Canada Day, and the hotel pays double time on holidays. What does she earn that day?

(b) A pipefitter's union agreement pays $42.00/h for the first 8 h in a day, time and a half for hours 9 to 12, and double time for every hour over 12. What does he earn for a 14 h day?

6. A courier is paid by the package: $3.10 each for the first 100 packages in a week, and $3.60 each for every package after that.

(a) This week she delivers 136 packages. What does she earn?

(b) She worked 28 h. What did she earn per hour? Is that more or less than the Alberta minimum wage?

(c) Give one advantage and one disadvantage of being paid by the piece.

7. Priya bids $3800 on a contract to paint the outside of a house. Paint and supplies cost her $640. She pays a helper $19.00/h for 24 h. She plans to work 40 h herself.

(a) What is her profit?

(b) What is her effective hourly rate?

(c) Rain slows the job, and Priya works 52 h instead (her helper's pay stays the same). What is her hourly rate now?

8. Find each person's earnings.

(a) Carmen sells furniture. She earns \$2100 a month plus 4% of all her sales. This month she sells \$38 750 of furniture.

(b) A real estate agent's company earns 7% on the first \$100 000 of a home's price and 3% on the rest. A house sells for \$485 000.

9. Jordan is offered two sales jobs. **Job A**: \$2400 a month salary. **Job B**: \$1200 a month plus 6% commission on all sales.

(a) Which job pays more in a month with sales of \$15 000?

(b) Which job pays more in a month with sales of \$28 000?

(c) What sales would make the two jobs pay the same? (Job B needs \$1200 more in commission than its base pay.)

(d) Give one advantage and one disadvantage of Job B.

10. Extra earnings.

(a) Rosa, Kai and Lin share \$840 in pooled tips by hours worked. Rosa worked 24 h, Kai 16 h and Lin 20 h. How much does each get?

(b) Rosa's wage is \$15.00/h. What are her total earnings for the week, with tips?

(c) A security guard earns \$19.25/h plus a premium of \$1.75/h for night hours. He works 40 h, and 16 of them are at night. What is his gross pay?

(d) A manager earns a salary of \$46 800 and gets a 3% year-end bonus. How much is the bonus?

11. Andre works 37.5 h at \$22.40/h. Use CPP 5.95%, EI 1.64% and income tax 16% of taxable income.

Earnings		**Deductions**	
Hours	37.5	CPP (5.95%)	
Rate	\$22.40	EI (1.64%)	
Gross pay		Income tax (16%)	
Taxable income		Union dues (before tax)	\$18.75
		Health plan	\$9.20
		Total deductions	
		Net pay	

(a) Complete the pay stub.

(b) What percent of his gross pay does Andre take home?

12. Each solution has a mistake. Find it and give the correct answer.

(a) Dev worked 6 h 45 min at \$16.80/h. He wrote: "$6.45 \times 16.80 = \$108.36$."

(b) Ty earns a salary of \$39 000 and is paid biweekly. He wrote: "$39\,000 \div 52 = \$750.00$ each pay."

13. Work backwards. Write the formula you use, and check your answer.

(a) Noor earns \$17.50/h at a car wash. How many hours must she work to earn \$630.00 gross?

(b) Noor takes home 81% of her gross pay. She wants to save \$900.00 of take-home pay for a laptop. How many hours must she work?

(c) A sales rep earns a straight 5% commission. What sales does she need to earn \$3000 this month?

(d) A summer job paid \$2268.00 gross for 144 h. What was the hourly rate?

Part B: Multiple Choice

14. Which job is most often paid by commission?

A. a tree planter **B.** a server **C.** a car salesperson **D.** a teacher

15. 4.6 h is the same as

A. 4 h 6 min **B.** 4 h 60 min **C.** 4 h 36 min **D.** 4 h 40 min

16. A salary of $41 600 a year is paid biweekly. Each pay is
A. $800.00 **B.** $1600.00 **C.** $1733.33 **D.** $3466.67

17. A worker earns $18.60/h. Her time-and-a-half overtime rate is
A. $20.10/h **B.** $37.20/h **C.** $9.30/h **D.** $27.90/h

18. In Alberta, Kim works 10 h a day from Monday to Thursday (40 h) and does not work Friday. How many overtime hours should she be paid?
A. 0 h **B.** 8 h **C.** 4 h **D.** 2 h

19. A salesperson earns 2% on the first $50 000 of sales and 4% on sales above $50 000. With sales of $80 000, he earns
A. $1600 **B.** $3200 **C.** $2400 **D.** $2200

20. Gross pay is $900.00. Deductions are CPP $53.55, EI $14.76 and income tax $135.00. Net pay is
A. $696.69 **B.** $1103.31 **C.** $765.00 **D.** $711.45

21. Gross pay = rate × hours. Which formula gives the number of hours?
A. hours = pay ÷ rate
B. hours = pay × rate
C. hours = rate ÷ pay
D. hours = pay − rate

Part C: Numerical Response

22. Numerical Response Sam's shift is from 6:30 a.m. to 3:15 p.m., with a 30 min unpaid lunch. The number of hours he is paid for, as a decimal, is _____ h.

23. Numerical Response Tomas earned $94.50 commission on a $2100.00 sale. His commission rate is _____%.

24. Numerical Response Lucas's gross pay is $720.00 and his total deductions are $172.80. The percent of his gross pay that he takes home is _____%.

25. Numerical Response Aiden earns $17.00/h. His gross pay last week was $697.00. The number of hours he worked is _____.
